

A Far Field Attractor in the Umbilicus Score of Volume Cartographer

Giovanni Pellerano

Milano, Italy

giovanni.pellerano@evilaliv3.org

Abstract—The umbilicus estimator of Volume Cartographer scores a candidate centre with a weighted sum of squared cosines divided by the sum of the weights, a weighted mean. We show that the limit of that mean far from the section is the largest eigenvalue of the normals’ covariance. Where this limit exceeds the score on the axis, the hill climb walks out of the volume. The published estimate leaves the grid on 135 of 360 slices, and 134 of those meet the condition. Without the division the score is a weighted sum, with a maximum at finite distance. The change is one line, now merged into the challenge’s code. On a run of 24 scrolls it keeps 576 of 576 slices inside the grid, against 350 as published. Only 2 of its four registered criteria pass, so the run does not show that the change generalises. Against the centroid of the sheet mask, by a margin fixed in advance, the sum wins on 8 of 10 third party references, out of sample. It costs a bias towards the denser side. The error of the hand references remains unmeasured, and we do not claim that the weighted sum is the right objective.

Index Terms—Herculaneum papyri, virtual unwrapping, umbilicus, objective functions, software defects, pre-registration.

I. Introduction

THE organisers of the Vesuvius Challenge name as their next goal, in their list of open problems for 2026, “making all of this reliable, on any scroll, without someone catching every failure” [1]. The scrolls of Herculaneum were carbonised by the eruption of Vesuvius and cannot be opened by hand. They are scanned in three dimensions, and software has to find the rolled sheet inside the scan. Spiral fitting [2] reconstructs that sheet by fitting the whole spiral at once, and it has one input that cannot be omitted: the umbilicus. That is the line through the centre of the roll, with one point per slice, a slice being one horizontal section of the scan at one height. The umbilicus is drawn by hand [3], the challenge publishes one for a small minority of its scrolls, and on the rest it has to be estimated before anything else can be attempted. A centre found without a person to check it removes one manual step, on every scroll, between a scan and a sheet that someone can read.

volume-cartographer [4], the toolchain of the challenge’s villa repository, contains an estimator for exactly this. It is the function `align_and_extract_umbilicus`, defined at `normalgridtools.cpp:15`, whose header comment calls it “a RANSAC-based approach to find a common umbilicus point”. It reads the normal grid, the published directions of the sheets of a scroll, perpendicular to the papyrus. It samples line segments from the grid and scores a candidate centre by how well the directions from the candidate to the segments line up with their normals. It then picks the best of a thousand random candidates and refines it with a

hill climb, a search that moves the candidate step by step towards a higher score. At the commit we measured, nothing in the repository calls the function but its own tests. Every line number in this paper refers to commit `23adee047`. At `2dcfaf6a0`, the tip of the default branch on 2026-09-17, the line at issue was still line 116. That tip is 50 commits later, and 0 of those commits touch the source file or its header.

We show that the score the function refines is a weighted mean whose limit far from the section can exceed its value on the axis. The hill climb then walks out of the scan (Section II). Removing the division turns the mean into a weighted sum, which has a maximum at finite distance. We measure the compiled function with and without the division against the fifteen scrolls that carry a hand drawn umbilicus, under verdicts written before the run (Sections III and IV). We also run it on the 23 scrolls of the competition set and PHerc0332, under four criteria registered before that run (Section V). We then measure what the change costs (Section VI) and whether the search shared the fault (Section VII). The patch (Section VIII) is that line, a seed argument that makes a run repeatable, and a regression test on synthetic sections whose centre is known. Pull request 1823 [5], which carries it, was merged into the main branch of villa on 21 September 2026 at 11:33:33 UTC.¹

The function, the normal grids and the reference umbilici are not ours. The first two are the challenge’s, and the references are the challenge’s and a third party’s (Section III). Ours are the analysis, the measurements, the patch and the tools that write every number. A reader can check each number without taking it on trust: the appendix names, for each section, table and figure, the file its numbers are read from and the tool that writes that file. The rules the verdicts were judged by were written and dated before the measurements [6]. They are in `src/prereg/`, and those of the run on the 24 scrolls and of the twenty seeds are in `src/inputs/`.

II. The defect

This section shows why dividing by the sum of the weights gives the score an attractor far from the scroll, and then tests that account on every slice that has a reference.

The refinement score runs over sampled segment midpoints, weights each by the inverse of its distance from the candidate and adds up the squared cosine between the segment normal and the direction to the segment. The last line of it divides (`normalgridtools.cpp:112-116`):

¹As the GitHub API gave it when read again on 28 September 2026 at 12:23:52 UTC.

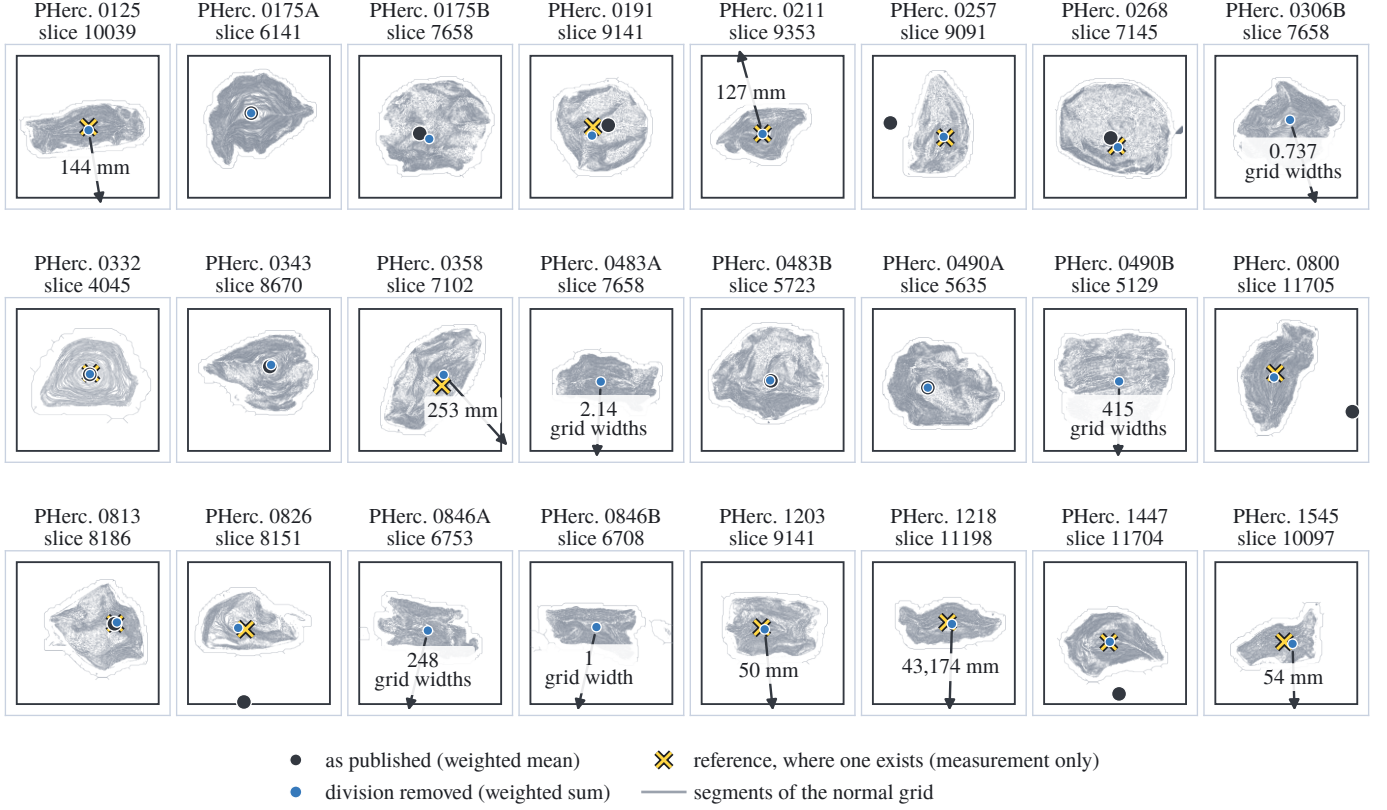


Fig. 1. One section of every scroll of the generalisation run, 24 panels sorted by scroll id. Grey: the segments of that scroll’s normal grid on the slice named in the title, with the bounds of the grid as a dark grey square; each panel is drawn in the units of its own grid, whose widths run from 3,941 to 12,145 pixels, with a margin of 8 % of the width on every side. Dark grey circle: the estimate of the published estimator, the weighted mean. Blue circle: the estimate with the division removed, the weighted sum. Yellow cross: the reference umbilicus at that slice, drawn only on the 14 scrolls that have one, published by the challenge on PHerc. 0125, PHerc. 0211, PHerc. 0332, PHerc. 0826 and annotated by a third party on PHerc. 0191, PHerc. 0257, PHerc. 0268, PHerc. 0358, PHerc. 0800, PHerc. 0813, PHerc. 1203, PHerc. 1218, PHerc. 1447, PHerc. 1545; the reference was used only to measure and never to choose anything, and the 10 scrolls PHerc. 0175A, PHerc. 0175B, PHerc. 0306B, PHerc. 0343, PHerc. 0483A, PHerc. 0483B, PHerc. 0490A, PHerc. 0490B, PHerc. 0846A, PHerc. 0846B have none. Dark grey arrow: the published estimate lies beyond the window, in the direction the arrow leaves the blue circle, at the labelled distance between the two estimates, in millimetres where the voxel size of the scan is on record and in grid widths otherwise; this happens on 11 of the 24 panels. On the slices drawn, the published estimate lies outside its grid on 11 panels, PHerc. 0125, PHerc. 0211, PHerc. 0306B, PHerc. 0358, PHerc. 0483A, PHerc. 0490B, PHerc. 0846A, PHerc. 0846B, PHerc. 1203, PHerc. 1218, PHerc. 1545, and the estimate with the division removed lies inside on every panel. No walk in this run was stopped by a cap: the estimates are those of the compiled function, which has no iteration limit, and every walk stopped by itself. Slice rule: of the 24 slices the run sampled on each scroll, evenly spaced along z , the panel shows the twelfth in order of z , the lower of the two that straddle the middle of the sampled band; no slice was chosen by looking at it. Over the whole run, 576 of 576 estimates fall inside the volume with the division removed, against 350 of 576 as published. Drawn by tools/figure_umbilicus_gallery.py from evidence/cpp-positions-24.csv, the cached grids of the run and the reference files; no estimate was recomputed.

```
float weight = 1.0f / std::max(100.0f, dist);
score += (cos_angle * cos_angle) * weight;
wsum += weight;
...
return score/wsum;
```

Without that division the quantity is a weighted sum. Let D be the distance of the candidate from the section and $\mathbf{u}$ the unit vector from the candidate towards it. Every weight tends to $1/D$, so the sum tends to zero: a positive continuous function that vanishes at infinity attains a maximum at finite distance, by construction. With the division it is a weighted mean. Numerator and denominator vanish together, and every direction from the candidate to a segment tends to $\mathbf{u}$. The limit far from the section, which we call the far field, is then the plain mean of the squared cosines over the N sampled normals $\mathbf{n}_i$,

$$\lim_{D \rightarrow \infty} \frac{\sum_i w_i \cos^2 \theta_i}{\sum_i w_i} = \frac{1}{N} \sum_{i=1}^N (\mathbf{n}_i \cdot \mathbf{u})^2 = \mathbf{u}^T C \mathbf{u}, \quad (1)$$

with $C = \frac{1}{N} \sum_i \mathbf{n}_i \mathbf{n}_i^T$. Every normal contributes equally to that limit, however far it is, because a mean does not know how many normals are contributing or where they are. On a complete circle of radial normals the limit is $1/2$ for every $\mathbf{u}$. On a partial section whose normals are coherent it rises towards 1 along the mean normal. It can exceed the score at the true centre once the normals there are noisy, so the mean stays high far away while the sum collapses (Fig. 2).

The maximum of that limit over $\mathbf{u}$ is the largest eigenvalue of C . That turns the observation of Fig. 4 into a criterion computable from the normals alone: the defect strikes partial predictions with coherent normals and spares complete sections. The criterion can be measured on the samples the estimator itself draws, 10,000 of them.² On the PHerc0826 slice of Fig. 4 the largest eigenvalue is 0.65, while the weighted mean at the published control point is 0.58. The far field is

²tools/far_field_eigenvalue.py, which writes evidence/far-field-eigenvalue.csv.

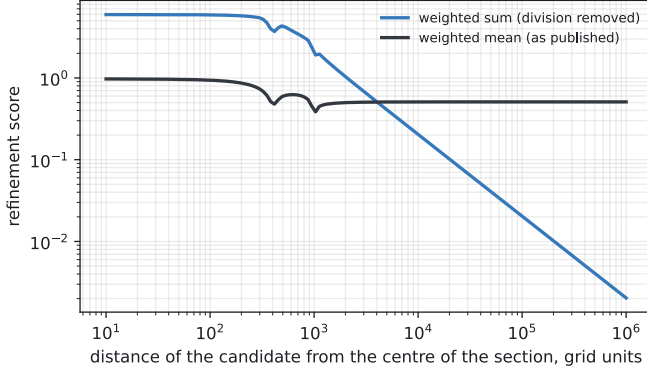


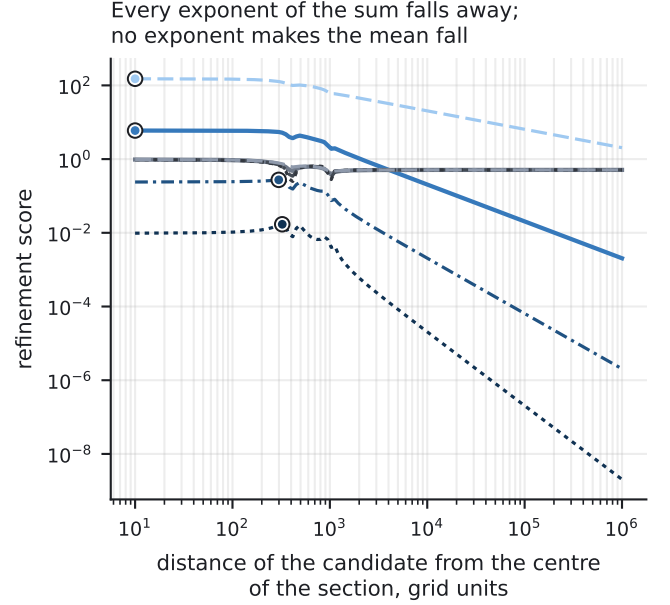
Fig. 2. Refinement score of `normalgridtools.cpp:98-117` on a synthetic section, one turn of a spiral of 4000 sampled points, against the distance of the candidate from the centre of the section. Weighted sum and weighted mean, same section, same samples. Recomputed by `tools/synthetic_density.py` on this section and on a concentric circle (`evidence/mechanism-curve.csv`): a million units away the sum retains $3.4\text{e-}4$ of its maximum and the mean 0.52; on the circle, $9.5\text{e-}4$ and 0.50.

therefore 1.11 times the score at the centre, and the climb has somewhere better to go. A synthetic full circle gives 0.50 against 1.00 at the centre. A half circle of radial normals still gives 0.50, because normals spanning half a turn cover both directions of the plane as evenly as normals spanning the whole of it. That is why the maximum of the mean sits at the true centre on the half circle of Table VI, 0 units away against 261 for the sum. The one section of that table where it does not is the ellipse, where it sits 6 units away, against 125 for the sum. It is also why the one complete concentric section of the primary block, PHerc0332, gives the same median with and without the division, 0.35 against 0.37 mm in Table III. What changes there is the count of runs outside the grid: 20 of 480 as published, against 0 with the division removed.

A faster falling weight does not rescue the mean, because a pure power cancels. With weights $1/d_i^p$ the factor D^{-p} stands in the numerator and in the denominator alike and leaves equation (1) as it is. The sum keeps the factor and vanishes as D^{-p} for every $p > 0$. A larger exponent moves the distance at which the plateau becomes visible; it does not lower the plateau (Fig. 3), and Section VIII-A measures the exponent on the fifteen scrolls.

The two objectives therefore fail on orthogonal properties. The mean fails when the normals of a section are coherent, which is what the eigenvalue measures. It is exact when they cover the plane however one-sided the section is, as the half circle shows. The sum fails when the density of the segments is asymmetric, which is what Section VI measures, and coherence does not trouble it. A real prediction is usually both at once, partial and one-sided, so neither objective is right in general, and our choice is made for the sections this toolchain actually processes.

Equation (1) speaks about every slice, so we put it to every slice that has a reference, the hand drawn umbilicus of Section III. An estimate leaves the grid when it ends outside the bounds of the scan. That is 360 slices on the 15 scrolls with a manual reference, the 24 slices of each scroll that Section III measures, run with the same 20 seeds per slice. For each slice



Filled circle: the largest value of each sum on the range drawn. The four means lie on one plateau, which is $\lambda_{\max}$ of the normals' covariance and does not depend on p ; the four sums leave that plateau at a rate p , each on its own units.

Fig. 3. The same score against the same distance as Fig. 2, with the exponent of the weight set to 0.5, 1, 1.5, 2. The four weighted means end on one plateau, the largest eigenvalue of C , which does not depend on the exponent. The four weighted sums leave it downwards with slope p . Drawn by `tools/figure_exponent.py` from `evidence/exponent-ablation-mechanism-curve.csv`; the rate at which each curve approaches or leaves the plateau is measured in `evidence/exponent-ablation-cancellation-rate.csv`.

we computed two things on the samples the table's own seed draws: the largest eigenvalue of C , and the weighted mean at that slice's reference control point.³ The reference serves only to evaluate the score at a point known to lie on the axis, which is measuring and not choosing.

Table II gives the answer. The bound exceeds the score on the axis on 192 slices. The published estimate leaves the grid on 135, and 134 of those satisfy the condition. Of the 168 slices where the bound stays below the axis, 1 walk out. The condition is therefore necessary on all but 1 of the 135 slices that leave, and not sufficient. Its sensitivity is 0.99 and its specificity 0.74. The 58 slices that satisfy it and stay inside are those where the climb does not reach the border from its random start, which the criterion does not claim to predict.

The outcomes here are those of the shipped walk, from the same 20 seeds per slice as the tables. The eigenvalues come from the grids and are the same under either instrument. Against the transcription's outcomes on the same slices, 3 slices change side under the majority rule. They are PHerc. 0257 z 8479, PHerc. 0268 z 2680, PHerc. 1203 z 5019. One

³By `tools/far_field_eigenvalue.py`.

TABLE I

Equation (1) with the scroll as the unit, the 24 slices of each of the 15 scrolls with a manual reference. Flagged: slices where the far field bound exceeds the weighted mean at the reference control point. Left: slices whose estimate as published leaves the grid, by the rule of Table II. Missed: leaving runs the condition did not flag. Generated from `evidence/far-field-per-scroll.csv`.

scroll	flagged	left	flagged and left	missed
PHerc0125	22	22	22	0
PHerc0211	17	13	13	0
PHerc0332	1	1	1	0
PHerc0826	24	16	16	0
PHercParis4	8	5	5	0
PHerc0191	3	0	0	0
PHerc0257	9	3	2	1
PHerc0268	10	1	1	0
PHerc0358	12	10	10	0
PHerc0800	10	6	6	0
PHerc0813	8	2	2	0
PHerc1203	10	8	8	0
PHerc1218	23	23	23	0
PHerc1447	18	11	11	0
PHerc1545	17	14	14	0

TABLE II

Equation (1) against the runs, on the 360 slices of the 15 scrolls with a manual reference. Rows: whether the far field bound exceeds the weighted mean at the reference control point of that slice. Columns: whether the estimate as published leaves the grid, by the rule fixed before the run, more than half of the 20 seeds landing outside. Generated from `evidence/far-field-prediction.csv`.

	leaves the grid	stays inside
$\lambda_{\max} > \text{mean at the reference}$	134	58
$\lambda_{\max} \leq \text{mean at the reference}$	1	167

of them is the miss. On PHerc. 0257 z 8479 the bound is 0.909 of the score on the axis, so the condition does not fire, and the published walk leaves the grid on 14 of 20 seeds under the shipped function against 10 of 20 under the transcription, on either side of the majority. Under the transcription the condition missed 0 slices; under the shipped function it misses 1. That is one slice at the majority line, and it is printed as a difference and not reconciled.

Those 360 slices are not independent observations. They are 24 from each of 15 scrolls, and slices of one scroll share a geometry, a surface prediction and a scanner, so a test that pools them would answer a question nobody asked. With the scroll as the unit no pooling is needed. On 14 of the 15 scrolls, not one run that left the grid escaped the condition. Runs left the grid on 14 scrolls, and on 13 of them every leaving run was flagged (Table I). The remaining scroll, PHerc0191, has no run to flag. To test the pairing, we shuffled the predictor among the slices of one scroll, independently in each scroll, which holds every scroll's own two counts fixed and breaks only the pairing between them. The statistic is a sum over scrolls of independent hypergeometric counts, so its tail under that null is exact by convolution: $p = 5.29\text{e-}29$. A simulation agrees. Of 200,000 shuffles, 0 reached the observed count, so $p < 5.0\text{e-}6$.

The hill climb at `normalgridtools.cpp:119-142` is therefore not searching for a maximum that is hard to find. On most real sections it is searching for one that is

not there, and it keeps stepping outwards because every step improves the mean a little. Nothing stops the walk: `normalgridtools.cpp:128` adds the step to the current best with no clamp to the grid. The answer can leave the volume by an arbitrary distance, and the routine still returns it as an estimate.

Fig. 4 is one real slice scored both ways, on one scale. A weighted mean and a weighted sum do not have the same units, so both panels are divided by the score at the published control point, which is the same physical place in both. As published, the field rises towards one edge of the grid and is as high far from the sheets as it is at the control point. Its maximum sits on that edge and not on the sheets. The climb stops 42.97 mm from the control point, outside the grid. With the division removed the field collapses away from the sheets, the maximum sits inside the cloud of sheet segments, and the climb stops 5.07 mm from the control point. In both cases the hill climb finds what its objective tells it to find.

The other score the function computes does not repair this. The candidate selection of lines 78 to 88, which picks the seed for the refinement, is an unweighted sum of the same squared cosines. It carries no distance weight at all, so nothing in it localises a maximum, and on the slices where the published objective peaks at the border that one does too. Dividing by the number of samples would change nothing, while dividing by the sum of the weights moves the maximum, because that sum depends on where the candidate is.

What the argument guarantees is narrower than what the measurement finds. A weighted sum whose weights fall as $1/D$ has a maximum at finite distance for every section, of any shape, at any resolution. But that maximum is interior to the plane and not to the grid, and the argument bounds the objective and not the number of sweeps the search takes to reach it. That every estimate lies inside the volume, and that every walk stopped by itself, are empirical results: 576 of 576 in Section V. Where the maximum sits is not guaranteed either, and Section VI measures it.

III. Against fifteen manual references

This section measures the compiled function against the fifteen scrolls that carry a hand drawn umbilicus, and applies to it the verdicts that were written before the run.

A. What is measured

Fifteen scrolls carry a manual umbilicus, which we call a reference. Five were published by the challenge and form the primary block; ten were clicked by a third party with no stake in the result [7] and form the confirmation block. The two blocks are never pooled. The pre-registration, the rule of each verdict written and dated before the measurement it judges,⁴ fixed that structure and the criteria, together with the argument of Section II. It was written before the estimator ran on any of the eleven scrolls whose grids were still to be downloaded, and before any accuracy number beyond four already reported, on PHerc0125, PHerc0211, PHerc0332, PHerc0826. So 4 of

⁴prereg/preregistration-fifteen-references.md, written 2026-09-15 15:33:39Z.

PHerc. 0826 xy slice 8000, refinement score divided by the score at the published control point

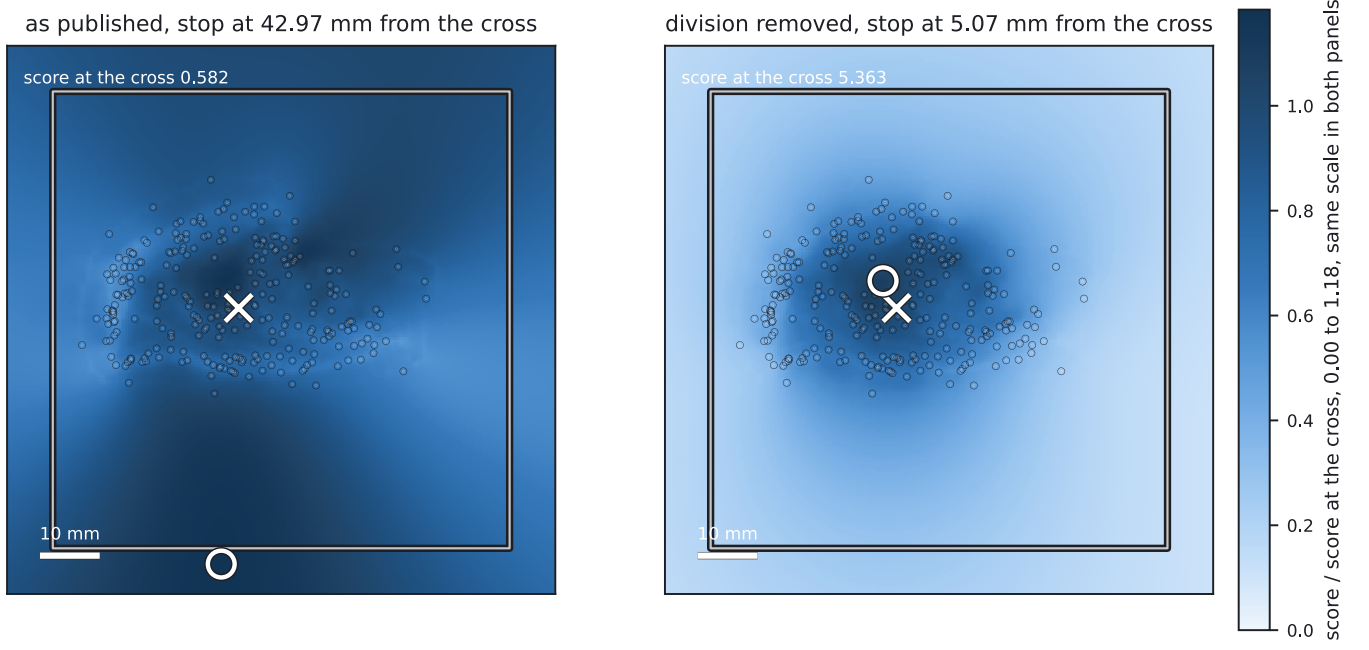


Fig. 4. Refinement score over PHerc. 0826, xy slice 8000, divided in each panel by the score at the published control point, so that both panels carry the same dimensionless scale. Left: as published, the weighted mean. Right: the weighted sum. Cross: the published control point. Circle: where the hill climb of `normalgridtools.cpp:119-142` stops. Rectangle: the bounds of the grid. Dots: one sampled segment midpoint in 80. Bar: 10 mm at 9.362 μm per voxel. Drawn by `inputs/figure_umbilicus.py` from that slice of the published normal grids and from the reference umbilicus of PHerc. 0826.

TABLE III

Primary block: the 5 umbilici published by the Vesuvius Challenge, one of them annotated by David Josey. Every rule is scored on the same 24 slices of the same scroll; the estimate is a polyline through those slices and the distance is taken at each of the scroll's control points, in millimetres of the scan the reference is annotated on, so the row is a median over 32 to 154 control points, not over slices. Bold is the estimator with the weighted sum, estimated with 20 seeds per slice: the median over the 20 per-seed medians, its 95 % percentile bootstrap interval in brackets (1,000 resamples, one seed drawn per slice and the control points resampled, generator seed 20260916). The margin is centroid minus estimator; $^{\circ}$ marks a margin whose interval includes zero, and † a margin smaller than 2.205 mm. That distance separates the hand annotation of PHerc. 1218 from a per slice centroid of the papyrus mask [8], not one hand from another, and it sits 0.06 mm from that scroll's own centroid column in Table IV, so the dagger reads as a margin smaller than the baseline's own distance from the reference, which is close to circular. It is kept because the pre-registration fixed the marking before any of these numbers were seen; Section IX gives the size of the overlap. It is a reading aid and not a criterion. The last column counts estimates outside the grid over all 480 runs, as published and with the division removed. Generated from `evidence/cpp-k20.csv`.

scroll	points	published	with the sum [95 %]	centroid	axis	margin [95 %]	outside
PHerc0125	75	136.59	1.59 [1.16, 2.15]	2.48	1.53	+0.90 [+0.33, +1.32] [†]	441 to 0
PHerc0211	84	63.63	1.37 [0.89, 2.14]	2.83	8.07	+1.46 [+0.68, +1.93] [†]	260 to 0
PHerc0332	154	0.35	0.37 [0.28, 0.43]	0.77	1.63	+0.39 [+0.33, +0.48] [†]	20 to 0
PHerc0826	45	53.13	4.29 [2.90, 5.70]	3.51	7.44	-0.78 [-2.20, +0.61] ^{◦†}	322 to 0
PHercParis4	127	4.36	1.16 [0.91, 1.52]	3.39	7.63	+2.23 [+1.87, +2.47]	103 to 0

the 5 scrolls of the primary block had been measured, with one seed, before the criteria were fixed. Of the 10 scrolls of the confirmation block, 0 had. The confirmation block was therefore out of sample and carries the accuracy claim, 8 of 10 below. The primary block carries the pre-registered structure and the count of estimates leaving the volume.

On each scroll the estimator runs on 24 slices evenly spaced along z , and the estimate is a polyline through them. The distance is read at every control point of the reference, in millimetres of the scan the reference is annotated on. The row of Tables III and IV is the median over those points, of which a scroll has 32 to 154. These are distances from a hand annotation, not errors: the annotation has an error of its

own, and a row that is closer is closer to one person's opinion of where the axis is. The baseline is the centroid, slice by slice, of the filled sheet mask, which is the simplest rule that uses the same input. A constant axis at the centre of the field is the control that says whether a scroll discriminates at all.

The tables are computed by calling the function itself, which we call the compiled function to tell it from the transcription, our copy of it in Python. `vc_gen_umbilicus`, the command line tool in `src/driver/`, which the merged change also adds to `villa`, reads a normal grid slice and calls `align_and_extract_umbilicus` with the seed argument. In all that is 14,400 calls: the 24 slices of each scroll, each with 20 seeds, under both objectives. They took 5.01 core

TABLE IV

Confirmation block: 10 manual references of a third party, *herculaneum-umbilici* [7], MIT, clicked on level 3 images at scale 8, one slice every about 480 in z, by someone who was not annotating for this purpose. A weaker reference by construction, kept in a block of its own and never pooled with Table III. Columns and marks as in Table III. Generated from *evidence/cpp-k20.csv*.

scroll	points	published	with the sum [95 %]	centroid	axis	margin [95 %]	outside
PHerc0191	62	6.26	4.31 [2.83, 5.71]	6.42	5.57	+2.11 [+0.71, +3.60] [†]	0 to 0
PHerc0257	40	9.34	1.73 [1.24, 2.76]	3.71	5.77	+1.97 [+0.95, +2.47] [†]	56 to 0
PHerc0268	44	9.38	4.44 [2.87, 5.83]	11.10	10.26	+6.66 [+5.27, +8.23]	30 to 0
PHerc0358	35	9.06	2.76 [1.58, 3.79]	4.90	6.26	+2.14 [+1.11, +3.32] [†]	200 to 0
PHerc0800	40	5.16	2.68 [1.77, 3.85]	6.63	9.78	+3.95 [+2.79, +4.87]	131 to 0
PHerc0813	48	10.46	7.29 [4.73, 9.26]	7.59	13.50	+0.30 [-1.67, +2.86] ^{o†}	40 to 0
PHerc1203	36	5.11	1.48 [1.00, 1.88]	4.38	4.48	+2.91 [+2.50, +3.38]	150 to 0
PHerc1218	36	101.98	2.39 [1.62, 3.63]	2.15	5.01	-0.24 [-1.48, +0.53] ^{o†}	459 to 0
PHerc1447	36	39.81	1.85 [1.23, 2.79]	3.18	6.73	+1.33 [+0.38, +1.95] [†]	235 to 0
PHerc1545	32	49.49	1.87 [1.33, 3.25]	2.40	7.05	+0.53 [-0.85, +1.07] ^{o†}	280 to 0

TABLE V

Two more rules, scored after the fact. The plateau point nearest the centroid and the maximum of the distance transform of the filled section, both from the umbilicus bench [9], on the same 24 slices, the same heights and the same control points as Tables III and IV and with the same metric; the first two columns repeat those tables so that a row reads across. Medians in millimetres with a 95 % percentile bootstrap interval over the control points (1,000 resamples, generator seed 20260916); the two rules are deterministic and take no seed. The margin is the best of the three simple rules on that scroll minus the estimator, positive where the estimator is the closer. ^t marks the 3 scrolls whose data set the plateau rule's threshold, where it is fitted and not tested. The blocks are those of Tables III and IV and are not pooled. Both rules are post hoc with respect to every pre-registration of this paper, written after the frozen tables were known: the pre-registered baseline criterion names the centroid, it was fixed before the run, and nothing in this table changes a pre-registered verdict. Generated from *evidence/baselines-k20.csv*.

scroll	with the sum [95 %]	centroid	plateau nearest centroid [95 %]	maximum of the distance transform [95 %]	margin
<i>Primary block</i>					
PHerc0125	1.59 [1.16, 2.15]	2.48	2.06 [1.71, 2.43] ^t	2.28 [1.88, 2.76]	+0.47
PHerc0211	1.37 [0.89, 2.14]	2.83	2.85 [1.84, 4.50] ^t	4.41 [3.16, 5.04]	+1.46
PHerc0332	0.37 [0.28, 0.43]	0.77	0.74 [0.61, 0.88]	1.07 [0.91, 1.25]	+0.37
PHerc0826	4.29 [2.90, 5.70]	3.51	2.78 [1.69, 4.41] ^t	4.61 [3.75, 5.33]	-1.51
PHercParis4	1.16 [0.91, 1.52]	3.39	3.29 [2.76, 3.87]	5.05 [4.46, 5.51]	+2.13
<i>Confirmation block</i>					
PHerc0191	4.31 [2.83, 5.71]	6.42	6.41 [5.65, 7.16]	6.63 [5.50, 7.29]	+2.10
PHerc0257	1.73 [1.24, 2.76]	3.71	3.58 [2.62, 4.19]	2.27 [1.77, 3.33]	+0.54
PHerc0268	4.44 [2.87, 5.83]	11.10	11.64 [10.06, 12.93]	12.58 [11.27, 13.60]	+6.66
PHerc0358	2.76 [1.58, 3.79]	4.90	5.22 [4.44, 7.27]	6.63 [5.27, 7.67]	+2.14
PHerc0800	2.68 [1.77, 3.85]	6.63	6.39 [4.90, 7.32]	5.86 [5.05, 6.32]	+3.18
PHerc0813	7.29 [4.73, 9.26]	7.59	7.59 [6.40, 9.05]	7.70 [6.38, 8.45]	+0.30
PHerc1203	1.48 [1.00, 1.88]	4.38	4.25 [2.36, 5.24]	3.13 [2.44, 4.11]	+1.65
PHerc1218	2.39 [1.62, 3.63]	2.15	2.24 [1.59, 3.09]	4.15 [3.23, 5.67]	-0.24
PHerc1447	1.85 [1.23, 2.79]	3.18	3.12 [2.28, 4.22]	3.52 [2.50, 5.78]	+1.27
PHerc1545	1.87 [1.33, 3.25]	2.40	1.88 [1.47, 2.71]	3.98 [2.33, 5.00]	+0.01

hours, 1.25 s for the mean call.⁵ Nothing bounded the search. The loop at `normalgridtools.cpp:119-142` halves its step only when none of the eight probes improves, it has no iteration limit, and it was left exactly so.

That recomputation gives the plainest number here. Each objective produced 7,200 estimates on the 15 scrolls with a manual reference. The published weighted mean leaves the grid 2,727 times; the weighted sum leaves it 0 times. Counted the other way, the mean is inside on 4,473 of 7,200. The sum is inside on 7,200 of 7,200. The totals hide how bad the worst rows are. On PHerc1218 the published objective keeps 21 of its 480 estimates inside the volume. On PHerc0125 it keeps 39 of 480. The weighted sum keeps every estimate of every scroll inside.

Every measurement this paper rests on comes from the compiled function, with no cap: the 15 scrolls of Tables III and IV, the exponent table and the run of Section V on 24

scrolls. Together they are 38,304 calls, and 0 of them met the wall clock guard. Six things still rest on the transcription, each named where it appears. Four need what the shipped function cannot be given. They are the search ablation of Table VII, the audit of the transcription's own cap below, the density bias on synthetic sections of real coordinates, and the post hoc bench run of Section IV. Two were computed on the transcription's medians of the fifteen scrolls and not recomputed: the rank tests of Section IV and the sector ratio correlations of Section VI.

A seed is the number that starts the random generator, and it is why every row carries an interval. The generator at `normalgridtools.cpp:51` is seeded from `std::random_device`, so two runs of one binary give two answers. On 4 saved slices, each run was compared with the median of 20 runs. With the division removed, the median distance between them is 0.05 to 1.31 mm, and the worst single run is 4.52 mm away. On the 2 slices where the published version leaves the grid, 0125-6891 and 0826-8000, it does

⁵Summed from the seconds of each call in *evidence/cpp-k20-estimates.csv*.

so on all 20 of 20 repeats of the compiled binary. With the division removed, not one repeat of any slice leaves it. Every estimate below is therefore repeated with 20 seeds per slice and reported with an interval, so that a margin can be read against the noise of the search. The seed argument this patch adds is what lets a row be repeated at all.

Until this revision the tables came from the transcription, and it was tied to the original where it counts: on candidates of a real slice the two compute the same score to machine precision. The compiled function comes from a stand-in build of the source at commit 23adee047, which reproduces both values the project's own unit test asserts. We compared the two row by row, under a criterion fixed before the run. A row agrees when each arm's median lies inside the other arm's 95 % interval and no mark on the row changes. By that criterion 28 of 30 rows agree. The column this paper claims, the weighted sum, agrees on 15 of 15. Its largest move is 0.18 mm, on PHerc0813, where the median goes from 7.11 to 7.29 mm. Under both instruments the estimate is inside the grid on all 480 runs of every scroll. The published column moves further, at most 3.57 mm on PHerc0125, which is what a column whose walk leaves the grid should do. No pre-registered verdict moves.

One count improved when the instrument changed, and we name it rather than absorb it: the number of scrolls in the confirmation block whose weighted sum loses to the centroid by at least 0.30 mm. The transcription reads it as 1 and the shipped function as 0. The scroll that moves is PHerc1218, whose margin is -0.32 mm by the one and -0.24 mm by the other, on either side of the line for a loss. That is a change of measuring instrument and not a new result. 2 rows disagree, PHerc0268, PHerc0813, both of the published column and neither one this paper counts as won. Their medians lie inside each other's intervals, and only the mark for a margin whose interval includes zero differs. The rule written before the run excused that case, and the tool that applied it does not. We left the tool stricter than its own declaration rather than edit it once it was known which rows it caught.

The transcription stops a walk at 2,000 sweeps, a limit fixed before any result was looked at, and only the published objective ever met it. Every capped walk was outside the grid already, so the cap shortens a runaway and flatters the published column. Rescoring the transcription's tables with the capped estimates carried 16 times further out⁶ moves 3 of 30 rows, all of them published ones and all away from the reference. The largest move is 0.87 mm, and the number of verdicts changed is none. The compiled run has no such cap. A wall clock guard of 600 s was declared before it and applied to the two objectives alike; of 14,400 walks, 0 met it, and the longest took 3.02 s. The shipped hill climb stopped by itself every time, on every runaway scroll, as the float distance and float candidate of `normalgridtools.cpp:106` and `normalgridtools.cpp:128` predict.

B. The verdicts, applied as written

This subsection applies the pre-registered verdicts exactly as they were written. Each block carries two criteria, repair

and baseline, with a threshold of 0.30 mm. Both were fixed in writing before the run and are applied without discretion to the medians over the 20 seeds.

In the primary block, removing the division gives a better median than the published function on 4 of 5 scrolls. It also brings the count of estimates outside the volume to zero on all of them, so repair is won. The estimator is closer to the reference than the centroid by at least the threshold on 4 of 5, and on no scroll is it worse than twice the centroid, so baseline is won. The confirmation block gives 10 of 10 on repair, again with no estimate outside the volume. Against the centroid it gives 8 of 10, so both criteria are won there as written. On 1 of those 8 rows, however, the interval does not support the margin, because it includes zero (PHerc1545). Going from the single-seed run to 20 seeds changed none of the verdicts. PHerc0813 is not counted among the wins: its margin is 0.2957 mm before rounding, which Table IV prints as 0.30 and which falls short of the threshold. Criterion C2 asks for six scrolls of ten,⁷ so the verdict of the confirmation block holds.

The interquartile range of the per-seed medians is at most 1.03 mm, on PHerc0813. The threshold is reached by 12 margins over the centroid. Of these, 1 have an interval that includes zero, 0 in the primary block (none) and 1 in the confirmation block (PHerc1545). Of the same margins, 8 are smaller than 2.205 mm, the distance on PHerc. 1218 between the third party's hand annotation and an umbilicus derived from published sheet instance labels. Section IX says what that distance is and what it costs the reading. Those margins are counted where the pre-registration counts them, because the criteria were written on point estimates and are not rewritten after the fact. The tables mark which of them the interval does not support and which fall below that distance.

One further count is descriptive, chosen after the fact, since the pre-registration fixed the estimates outside the grid and not a distance. It draws a line at 10 mm from the reference, a distance no disagreement between annotators approaches. The published function leaves 7 of 15 scrolls above that line, and the weighted sum leaves 0. Nothing is declared won by it. We report it because the worst row of the sum is 7.29 mm, so the line is cleared by a margin a reader can see rather than by a hair.

IV. Rules and readings scored after the fact

This section gathers what was scored after the pre-registered verdicts were known. None of it changes a verdict.

Two rules the tables above do not carry will occur to a reader who knows the code. The first is the maximum of the distance transform of the filled section, which is the rule the upstream mask based tool applies. The second is the plateau point nearest the centroid, the closest of the simple mask rules in the umbilicus bench [9]. Table V scores both on the slices, the heights and the control points of Tables III and IV, with the same metric. Both rules are post hoc. We read each scroll against the best of the three simple rules at the pre-registered threshold of 0.30 mm. The weighted sum is the closer on 4

⁶evidence/search-cap.csv.

⁷prereg/preregistration-fifteen-references.md, line 116.

of the 5 scrolls of the primary block and on 7 of 10 in the confirmation block. On PHerc0813 it is ahead of the centroid by 0.30 mm as printed, which falls short of the threshold before rounding and is counted as not won, as in Section III. The maximum of the distance transform is never the closer of the two: it wins on 0 of 15 scrolls. The plateau rule is closer on 2, PHerc0826, PHerc1218. Its threshold was chosen on the data of PHerc0125, PHerc0211 and PHerc0826, so on the first of those two it is fitted and not tested, and on 1 it is tested.

The umbilicus bench asks the same question on a different sample, and the two are not mixed. It scores 5 mask rules at level 3 of the published surface predictions, on 46 heights per scroll fixed by the store. We ran both variants of the estimator inside it on its 4 scrolls, with the transcription. Its rows cannot be set beside the rows above: on PHerc0826 the same centroid code reads 3.12 mm there and 3.51 on the slices of Table V, a difference of 0.39 mm, larger than the pre-registered threshold. What crosses from one sample to the other is that the loss falls on the same scroll and in the same direction. There the weighted sum reads 4.16 mm against 2.47 for the plateau point nearest the centroid, and 4 of the 5 mask rules are closer than it. The runaway crosses too. On PHerc0125, PHerc0211 and PHerc0826 the published function leaves 859, 540 and 568 of its 920 estimates outside the grid, and on PHerc0332 it leaves none. The weighted sum leaves 0 of 3,680 over the four.

The estimator does lose, and it is worth being exact about where. On PHerc0826 the centroid wins. The estimator's median there is 4.29 mm [2.90, 5.70], against 3.51 for the centroid. The margin is -0.78 mm [-2.20, 0.61], and its interval includes zero. It is called a loss because the pre-registered criterion reads the point estimate, and Section VI gives the mechanism that makes it the expected direction.

Against the best of the three simple rules of Table V, the estimator loses on 2 scrolls of 15. On PHerc0826 it reads 4.29 mm [2.90, 5.70], against 2.78 [1.69, 4.41] for the plateau point nearest the centroid. That gap of 1.51 mm is smaller than the 2.205 mm of PHerc. 1218, and it is taken by a rule whose threshold was fitted on that scroll. On PHerc1218 the estimator reads 2.39 mm [1.62, 3.63], against 2.15 [1.64, 2.94] for the centroid. That gap, 0.24 mm, is smaller than the same 2.205 mm.

On one scroll nothing can be read at all. On PHerc0125 the constant axis at the centre of the field is closer to the reference than either estimator, which is a fact about the scroll and not about any estimator. The distance of that control from the reference is 1.53 mm there, the smallest of the fifteen, against a median of 6.73 mm. A rule that returns the field centre scores that well while knowing nothing about the scroll, so a row orders rules only where they are closer to the reference than that control is. The weighted sum is closer on all 14 of the other scrolls, where the control runs from 1.63 to 13.50 mm.

Counting scrolls is a weak use of fifteen paired measurements, and it is the use the pre-registration fixed. The tests the literature prescribes were run afterwards, on the transcription's medians of the same scrolls,⁸ and they disagree. A Friedman test compares the 4 rules over 15 scrolls. It rejects the null

that they are interchangeable, χ^2 23.08 at p 3.9e-5. The mean ranks are 1.27 for the estimator with the division removed and 2.33 for the centroid. The constant axis ranks 3.07 and the published function 3.33. Pairwise Wilcoxon tests with Holm correction then separate the repaired estimator from the centroid at p 0.0034. The Nemenyi test does not separate them: the gap in mean ranks is 1.07, against a critical difference of 1.21. We cannot claim, then, that the weighted sum beats the centroid under every test the literature prescribes.

V. The run on twenty-four scrolls

Accuracy can only be measured where a reference exists. For the other scrolls we registered a run before its grids were downloaded,⁹ on the 23 scrolls of the competition set and PHerc0332, one of the scrolls with a challenge reference. PHercParis4, which also carries one, is not in it. The run is the compiled function, called through `vc_gen_umbilicus` with no cap, on 24 slices of each scroll, 576 slices in all. The registration names four criteria and requires all four before the change can be said to generalise.

The first criterion asks that the corrected estimate stay inside the grid on at least 23 of 24 slices of every scroll, and it passes. With the division removed the count is 576 of 576, every slice of every one of the 24 scrolls. As published, 350 of 576 estimates fall inside the grid. On 12 scrolls fewer than two thirds of the slices stay inside, and only 3 scrolls keep all of them. The transcription read 351 and 576 on the same slices and seeds, so the repaired count is the same under either instrument and the published one differs by one estimate. These counts are over 576 slices at 1 seed each.

The second criterion asks that the median spread between 3 seeds per slice of the weighted sum stay below 1 per cent of the grid width on every scroll. It fails on 1 scroll, PHerc0846A, at 1.18 per cent. The third asks that the median jump between consecutive slices stay below 2 per cent of the grid width on every scroll. It fails on 23 of the 24 scrolls, with a median jump of up to 9.62 per cent on PHerc0490B. The fourth asks for an interior maximum on at least 90 per cent of the scrolls, and it passes with 23 of 24. By its own registered criteria the run therefore does not show that the change generalises: it passes 2 of the four. What it shows is the first criterion, that the estimate stays inside the scan.

Fig. 1 is one section of each of those scrolls, chosen by a rule and not by looking, with both estimates and, where one exists, the reference. On the scrolls without a reference the figure shows where the estimate falls and that it stays inside the volume. It does not show that the estimate is right, and nothing here claims that.

VI. The cost

This section measures what the one line costs: the weighted sum leans towards the side of a section that holds more sheet.

That the weighted sum has an interior maximum says nothing about where that maximum is. A sum rewards the side of the section that contributes more segments, while a mean normalises the density away. We put the question to

⁸evidence/fifteen-k20.csv, read by tools/rev1_inference.py.

⁹inputs/preregistration-twenty-four-scrolls.md, written 2026-09-15.

TABLE VI

Synthetic sections with a centre known by construction, 8000 by 8000 grid units, radii 300 to 3500. Position of the maximum of each objective on a lattice of step 4, and of the full estimate with the search of `normalgridtools.cpp:78-142`, medians over 20 seeds, in grid units (1 unit = 9.362 μm). Error: distance from the true centre. Δx : offset along x , positive towards the side that holds more segments. The density rows keep every segment with x above the centre and one in 2, 5 and 10 of those below it; the half circle keeps the half with y above the centre, so its offset is along y and its error column carries it. Generated from `evidence/synthetic-density.csv`.

section	sum, maximum error		mean, maximum error		estimate	
		Δx		Δx	sum	mean
circle 1:1	47	-2	0	+0	46	0
ellipse 1.4	125	-36	6	+0	126	6
half circle	261	-12	0	+0	259	0
density 1:2	202	+200	0	+0	202	0
density 1:5	229	+226	0	+0	230	0
density 1:10	244	+242	0	+0	244	0

sections whose centre is known, with the prediction written beforehand in a note dated and marked post hoc with respect to the first version of this measurement.¹⁰ The note considered a full circle whose lower half is thinned to one segment in ten. It predicted that the maximum of the sum would move towards the dense side by 50 to 250 grid units, and the maximum of the mean by less than 10.

Table VI gives the answer. On the uniform circle the maximum of the sum sits 47 units from the true centre, and that of the mean 0. With the lower half thinned ten times, the maximum of the sum moves +242 units along x towards the dense side and sits 2.29 mm from the true centre. The maximum of the mean moves +0. The prediction for the sum held, and the prediction for the mean held. The bias grows with the ratio and is largest on the half circle, where one side has no segments at all. There the maximum of the sum sits 261 units from the centre, and that of the mean 0. The full estimate follows the maximum in every row.

On these sections the published version is the more accurate one. On concentric circles it lands 0.004 mm from the true centre, against 0.430 mm for the sum.¹¹ The maximum of the mean sits at the true centre on every section of Table VI but the ellipse. The sum pays for its guaranteed maximum with a bias towards density, of 0.44 to 2.44 mm on these sections. Those millimetres are distances from a centre known by construction, and they are not a term to be added to the distances of Section III. Those are measured from the estimator's own output against a hand annotation, so whatever bias a real section induces is already inside them. The widest of the synthetic biases comes from a section with one side empty, which no slice of a real prediction is.

We made the same measurement on the real predictions. The segments of the 24 slices of each scroll were counted in eight angular sectors around the reference axis, which serves only as the origin of the sectors. The sector ratio of a scroll is the count of its most populated sector over its least populated

one, median over the slices.¹² On the primary block the ratio orders the scrolls exactly as the estimator's distance from the reference does, a Spearman correlation of +1.00 over 5 scrolls. The one primary scroll where the centroid wins, PHerc0826, has the highest ratio of the fifteen: 6.38, against a median of 3.59.

That ordering does not survive the fifteen. Over all of them, the Spearman correlation between the ratio and the distance from the reference is +0.33 (p 0.23). Between the ratio and the margin over the centroid it is -0.35 (p 0.20). The two confirmation scrolls where the centroid wins or nearly wins, PHerc1218 and PHerc0813, are not the one-sided ones by this measure. Their ratios are 4.08 and 3.59. Counted from the least one-sided, they rank 10 and 8 of 15. The prediction that they would be asymmetric is therefore not confirmed, and we offer no second measure of one-sidedness to rescue it. On the real scrolls the density bias explains the one primary loss and not the ranking. The one line trades an unbounded failure for a bounded bias towards the side of the section the prediction happens to have found. On a very one-sided prediction that bias can exceed the distance of the centroid baseline.

VII. The objective, not the search

This section asks whether the search was at fault too, and finds that the defect was the objective.

Our first assumption was that the search was at fault as well, and we rewrote it. The rewrite used double precision throughout and confined the walk to the volume. Its step halved when none of the eight probes improved, instead of chaining moves inside the 3 by 3 loop, and it had one objective instead of two. Table VII measures the four combinations on the same slices and seeds as Section III.

Take the weighted mean first. The rewritten search keeps 7,200 of 7,200 estimates inside the grid, against 4,484 for the published walk, but only because it is clipped there. It still lands 10 mm or more from the reference on 8 of 15 scrolls. At worst it is 39.97 mm away. The published walk does much the same, landing that far on 7 scrolls and at worst 133.02 mm away. With the weighted sum neither search fails on any scroll, and their worst rows are 7.11 and 6.50 mm. The rewritten search beats the centroid by the pre-registered margin on 4 of 15 scrolls with the mean. With the sum it does so on 13, against 13 for the one line. In the median it is better than the one line on 11 of 15 scrolls with the sum, and on 1 with the mean. A well posed objective makes even a loose search converge. The estimates outside the volume disappear without a clamp, because the score falls away in every direction and no step outwards keeps improving it.

Column (d) is closer on most of the scrolls, and the size of that difference is what decides the patch. Against the one line it is further on 3 scrolls and level on 1. Its median gain is 0.06 mm. It gains at most 0.61 mm and loses at most 0.51. Over the fifteen, the medians are 1.84 mm for column (d) and 1.87 for column (b). Only 1 of the 15 scrolls reaches the pre-registered 0.30 mm in its favour (PHerc0813). In the other

¹⁰`prereg/note-density-bias.md`.

¹¹`evidence/synthetic-sections-refit.csv`.

¹²`evidence/density-asymmetry-fifteen.csv`; the distances and margins it is set against are the transcription's, `evidence/fifteen-k20.csv`.

TABLE VII

The four cells of the ablation on the fifteen scrolls, median distance from the reference in millimetres, 20 seeds per slice, with the 95 % bootstrap interval of Section III. (a) as published: weighted mean, the search of `normalgridtools.cpp:78-142`. (b) the weighted sum, search untouched: the one line. (c) the search rewritten (double precision, walk clipped to the grid, eight probes from a fixed centre and a step that halves when none improves, one objective for seed and refinement), weighted mean. (d) the rewritten search, weighted sum. This table is the transcription's, columns (a) and (b) included. An ablation is measured on the instrument that can ablate the thing, and the shipped function has no replaceable search: columns (c) and (d) replace it, which is a property of `align_and_extract_umbilicus` and not a limitation of this work. Columns (a) and (b) are therefore the same quantity as Tables III and IV read by the other instrument, and they differ on 26 of 30 cells, by at most 3.57 mm. Before the tables were recomputed on the shipped function they matched row for row, which is what the two columns were built to do. Generated from `evidence/ablation-search.csv`.

scroll	(a) weighted mean, as published	(b) weighted sum, the one line	(c) rewritten search, mean	(d) rewritten search, sum
PHerc0125	133.02 [107.99, 190.24]	1.59 [1.15, 2.17]	39.96 [39.50, 40.60]	1.64 [1.18, 2.21]
PHerc0211	63.81 [45.21, 95.60]	1.47 [0.98, 2.38]	37.97 [33.43, 40.24]	1.32 [0.85, 1.97]
PHerc0332	0.35 [0.28, 0.41]	0.36 [0.30, 0.42]	0.35 [0.28, 0.41]	0.36 [0.30, 0.42]
PHerc0826	53.21 [40.75, 66.12]	4.27 [3.03, 5.44]	39.97 [37.27, 41.35]	4.30 [2.81, 5.37]
PHercParis4	4.25 [1.47, 26.57]	1.13 [0.83, 1.48]	3.12 [1.37, 11.04]	1.07 [0.81, 1.41]
PHerc0191	6.28 [4.39, 8.50]	4.15 [2.64, 6.03]	6.22 [4.56, 8.23]	4.66 [3.21, 6.02]
PHerc0257	8.37 [2.75, 31.17]	1.87 [1.26, 2.72]	5.43 [2.17, 30.40]	1.84 [1.28, 2.73]
PHerc0268	8.27 [5.63, 11.05]	4.32 [2.69, 5.87]	8.18 [5.24, 10.90]	4.30 [2.79, 5.95]
PHerc0358	8.46 [3.92, 97.85]	2.89 [1.74, 3.86]	10.38 [3.70, 37.37]	2.75 [1.63, 3.87]
PHerc0800	5.29 [3.49, 36.18]	2.68 [1.73, 3.83]	4.80 [3.30, 33.00]	2.66 [1.63, 3.70]
PHerc0813	10.49 [7.10, 12.45]	7.11 [4.42, 9.33]	10.26 [7.62, 12.29]	6.50 [4.31, 8.69]
PHerc1203	4.80 [1.46, 34.91]	1.44 [0.98, 1.77]	2.85 [1.40, 16.83]	1.38 [1.01, 1.75]
PHerc1218	104.09 [82.99, 150.90]	2.47 [1.66, 3.89]	33.27 [29.88, 35.68]	2.24 [1.60, 3.62]
PHerc1447	38.57 [28.75, 55.24]	1.76 [1.15, 2.56]	33.20 [25.73, 38.13]	1.69 [1.17, 2.51]
PHerc1545	49.55 [17.68, 68.47]	1.87 [1.24, 3.38]	32.66 [9.14, 40.16]	1.80 [1.25, 3.19]

direction, 1 reaches it (PHerc0191). By the rule the rest of this paper decides on, the two are not separated.

Column (c) settles what the rewrite repairs on its own. With the mean it still lands 10 mm or more from the reference on 8 scrolls, and it beats the centroid on 4 of 15, where the line does so on 13. So the search was not the defect. The rewrite is four changes to a search and a new one to review. It buys a median of 0.06 mm where the objective is right, and it repairs nothing where it is wrong, which is the argument for one line rather than for a rewrite.

VIII. The patch

This section describes the change as it was merged and what each of its parts is for.

The change was merged as commit 8839ed689, the squash of pull request 1823.¹³ It touches 6 files: the function and its header, the command line tool `vc_gen_umbilicus` and its build entry, the test file, and a fixture slice of PHerc. 0125. Our measurements were not made on that commit but on the stand-in build of Section III, which compiles the function as changed by 7a4129ald, the head of the pull request on the fork before the squash. Line 116 returns `score` instead of `score/wsum` and the accumulator `wsum` goes. The comment above the return states the trade of Section VI rather than presenting the change as a repair without cost. It says that the sum favours candidates near many segments as well as candidates the normals point at. It says that on synthetic sections with a known centre the mean is exact and the sum is biased towards the denser side. It also says that on the fifteen real scrolls the failure persists with the walk clipped to the grid.

The line, the seed argument and the tests answer different questions and should be read separately. The line changes the objective and carries the measured claim of this paper. The

seed argument makes a run repeatable and claims nothing about the objective. The synthetic tests bound the bias the line introduces, and the test on a real slice guards the line itself.

The generator that draws the 10,000 sampled segments and the 1,000 candidate points was a static `std::mt19937` seeded from `std::random_device`, so no run could be repeated. The signature gains one argument with a default:

```
cv::Vec2f align_and_extract_umbilicus(
    const GridStore& grid_store,
    std::optional<std::uint32_t> seed = std::nullopt);
```

With no seed the behaviour is exactly as before, the one static generator seeded once from the hardware; with a seed a fresh generator is seeded with that value and the run repeats exactly. The function has no other caller in the tree, so the change is source compatible.

The merged change adds 5 cases to the function's own test file, `test_normalgridtools.cpp`. One asserts that two calls with the same seed return the same point. Three more are on `GridStore` sections built as in Table VI: concentric circles, a half circle, and circles thinned 1:10 on one side. Each is estimated with a fixed seed and asserted to lie inside the grid and near the known centre, within 1.0, 5.0 and 4.5 mm respectively. The thinned section also asserts that the estimate leans towards the dense side. The tolerances are derived from the table, not from a run. Each is at least 1.56 times the p90 of the estimate's error on that section over 20 seeds, so that no seed fails them by statistics alone. Those p90 values are 0.61, 3.20 and 2.71 mm. The fifth case runs the estimator on the fixture slice of PHerc. 0125 with five seeds and asserts that the estimate stays inside the grid and near the challenge's umbilicus of that scroll. Its own comment reports that the weighted mean leaves that slice on every seed, so unlike the other four it fails if the division returns. Before the change the project's suite ran 17 of the 83 lines of this function (20.48 per cent). It reached neither the objective nor the search, so nothing in it could have disagreed with the division.

¹³`evidence/merged-patch.csv`, written by `tools/merged_patch.py` from a clone of villa.

The test file of 7a4129a1d holds the first four cases. We built it from that commit’s own sources, against OpenCV core and imgproc and without Qt, with flags and an OpenCV that are not the project’s image; neither touches the arithmetic under test. All 15 cases of the file pass. On the concentric circles the new cases put the estimate 0.23 mm from the known centre, against a tolerance of 1.0. On the half circle the estimate is 2.13 mm away, against 5.0. On the thinned circles it is 2.02 mm away, against 4.5. That last estimate sits at x 4206, on the dense side of the centre as Section VI predicts. Every estimate is inside the grid, and the repeated seed gives the same point to the bit. A second run in a fresh process gave the same 5 estimates.¹⁴ We did not run the fifth case. The four synthetic cases cannot tell the sum from the mean, because on these sections the mean is exact. They bound the sum’s known bias so that it cannot grow unnoticed, and they check that the walk settles inside the grid.

A. The exponent of the weight

The patch leaves the exponent of the weight as it was, and this subsection measures whether it should. The exponent of $1/\max(100, d)$ came with the function, and keeping it is a choice to be measured rather than assumed. We reran the estimator over the 15 scrolls of Section III, on the same slices and seeds, with nothing changed but the exponent, which took the values 0.5, 1, 1.5, 2. The row $p = 1$ is not recompiled: it is the weighted sum of the main run, the shipped expression itself, so Table VIII can be read beside every other number here. In the transcription’s run of the same exponents, which did compute $p = 1$, that row reproduced the transcription’s own estimates with 0 differences over 7,200.¹⁵

Staying inside the grid does not choose an exponent, and accuracy does. Every exponent tried keeps all 7,200 estimates inside the grid, against 2,727 outside for the published mean. That is the algebra of Section II again: the repair comes from not dividing, not from the value of p . We measured the rate at which each curve approaches its limit too, on 3 sections and 5 exponents up to $p = 4$, at 4 distances a decade apart out to 10^9 units. The departure of the mean from $\lambda_{\max}$ divides by 8.7 to 10.5 a decade, which is $O(1/D)$, while the sum divides by 10^p . The bar fixed before that run is met on all 15 of 15 combinations (Fig. 3). Accuracy is another matter. The median over the fifteen scrolls is lowest at 1,¹⁶ which is interior to the range tried rather than at its edge. Meanwhile the declared cost grows with the exponent on both fronts at once, the density bias and the spread between seeds.

Those medians are not separated by their intervals. At 1 the interval is [1.59, 2.76] mm. Of the exponents tried, only $p = 2$ lies clear of it. 0.5 and 1.5 overlap it, so the aggregate alone does not put 1 ahead of them. The aggregate also throws away the pairing, and the pairing is what the rest of this paper decides on. Scroll by scroll, with the pre-registered threshold of 0.30 mm, each exponent was compared with 1. None is better on more than 1 of the 15 scrolls, the one that reaches it

TABLE VIII

The exponent of the weight on the 15 scrolls with a manual reference, 24 slices and 20 seeds each, everything but the exponent as in Table III. Outside: estimates falling outside the grid. Median: the median over the 15 scrolls of the per scroll median distance from the reference, with a 95 % percentile bootstrap interval (1,000 resamples, generator seed 20260916). The unit resampled there is the scroll and not the control point, because the quantity is a median over scrolls; the arithmetic is that of Tables III and IV and only the unit differs. That median and its interval run over the two blocks together, as this row did before the interval was added. Pooling is admissible here because no pre-registered verdict of this paper is taken from a pooled sample. Beats: scrolls where the margin over the centroid reaches the pre-registered 0.30 mm. Spread: the median spread between seeds. Bias: the density bias on the section thinned ten times, as in Table VI. The last row is the published weighted mean on the same runs. Every row here is the compiled function, called through `vc_gen_umbilicus`, but three of the five rows are compiled variants of an objective upstream does not ship: the weight at `normalgridtools.cpp:112` is written `1.0f / std::max(100.0f, dist)` with no exponent, so p of 0.5, 1.5 and 2 are one line edits of it, built by a substitution anchored on that exact line which fails the build rather than produce a binary nobody can characterise. So on those three rows the compiled figure carries no more authority than the transcription’s did; what it buys is that p equal to 1, which is the shipped expression itself, and the control row, which is the shipped weighted mean, now agree with Tables III and IV to the digit instead of reading 8.46 mm against 9.38 mm for one quantity on one page. The bias column is the one thing in this table still measured by the transcription: its sections are polylines of real coordinates and the shipped `GridStore` holds integer points, so the shipped function cannot be given them without rounding them first. Generated from `evidence/cpp-exponent-fifteen-k20.csv`, `evidence/cpp-exponent-aggregate.csv` and their companions.

p	outside	median mm	beats	spread mm	bias mm
0.5	0 of 7,200	3.38 [2.09, 5.54]	10 of 15	0.34	1.80
1.0	0 of 7,200	1.87 [1.59, 2.76]	12 of 15	0.57	2.29
1.5	0 of 7,200	2.44 [2.23, 3.08]	12 of 15	1.04	4.48
2.0	0 of 7,200	4.99 [3.50, 6.01]	5 of 15	3.20	5.62
mean	2727 of 7,200	9.38 [5.16, 49.49]	3 of 15	–	0.00

being 1.5. Each is worse on 8 to 13. That is the comparison the exponent is left at 1 on. It is a reason not to move, rather than a demonstration that 1 is right.

One reading of that table can be turned against the choice. The smallest exponent, 0.5, passes three of the four bars written before the run, in a declaration this folder does not carry. It keeps every estimate inside the grid too. Its spread between seeds is smaller, 0.34 against 0.57 mm. Its density bias on the thinned section is smaller too, 1.80 against 2.29 mm. It fails on the fourth bar, accuracy. The exponent 0.5 is better on 0 scrolls and worse on 10. A reader who puts the density bias ahead of accuracy would prefer it. The rule that puts accuracy first was written before any of these numbers were seen, and we apply it as written.

IX. Limits

This section sets out what the measurements above do not establish.

They do not establish that the weighted sum is the right objective. On complete concentric sections the mean is the more accurate one, by the margin of Section VI, and a reader working on such sections should keep it. The division is wrong on the partial, one-sided predictions this toolchain actually processes, where it lets the search leave the volume. The claim is made in that narrower form.

They do not establish accuracy, because the error of the references is not known. The distances of Section III are distances

¹⁴`evidence/regression-test-run.csv`.

¹⁵`evidence/exponent-ablation-k20-estimates.csv` against `evidence/k20-estimates.csv`.

¹⁶The medians are 3.38, 1.87, 2.44 and 4.99 mm at p of 0.5, 1, 1.5, 2.

from hand annotations whose own error is not measurable on any scroll of today's public data. The 5 challenge umbilici we score and the 10 of the third party fall on 15 different scrolls, one hand each, and the two sets do not meet. The challenge publishes one more, for PHerc0139, again one hand, which we do not score because no surface prediction exists on the scan it is drawn on.¹⁷ What PHerc. 1218 carries beside its hand annotation is not a second hand. It is an umbilicus derived from published sheet instance labels [8], by a script that takes the centroid of the papyrus pixels of each slice and reads the labels only as a mask. The second reference is therefore the construction we score as the centroid baseline. The 2.003 to 2.205 mm that separate the two on that scroll are a hand against a mask centroid. Of the 12 margins won, 8 are below that distance, and what they are below is the baseline's own error rather than the error of a hand.

The same derivation runs from the masked scan and the surface prediction the challenge publishes, so we repeated it on every scroll of Section III.¹⁸ That gives 15 pairs, each a hand against a mask centroid. The distance runs from 0.766 mm on PHerc0332 to 10.748 mm on PHerc0268, a ratio of 14.0 between the largest and the smallest. Its median is 3.660 mm, with quartiles 2.681 and 5.942. PHerc. 1218 is the second smallest of the 15, and 13 lie above 2.205 mm, so that number is not a scale for the others and we do not use it as one. Over those scrolls the distance tracks the centroid column of Tables III and IV, with a Pearson correlation of 0.9954 and a median difference of 0.19 mm. That is the overlap the dagger of Table III inherits: the two are close to the same quantity, and no criterion of this paper rests on either.

They do not establish that the function can warn of its own runaway. The far field bound is a byproduct of the samples the function already draws, but Table II compares it with the weighted mean at the reference control point, which the code does not have. Compared instead with the refined score, the only score the function holds when it returns, the same test fires on 21 slices of 360. Of those, 4 are on a run that stayed inside. It misses 119 of the 136 runs that left on that single seed, a count taken on one seed and not by the majority of Table II. The sensitivity is 0.12, against 0.99 for the comparison at the reference. Once the walk has reached the far field its own score has risen to the bound, so the comparison that was true where the search started is false where it stops.

Nor does the bounded walk rest on a guarantee. With the sum the score falls away in every direction, and on these runs the walk settles inside the grid every time, but that is measured, not proved.

X. Open questions

How large is the error of a hand drawn umbilicus? Every distance in this paper is taken from one hand, and no published scroll carries two, so the margins of Section III cannot be set against the disagreement between annotators. The one second reference we have, on PHerc. 1218, turned out to be a mask centroid (Section IX). Two independent hand annotations of

the same scroll would settle it, and they would give the scale on which a margin of 0.30 mm can be read.

Can the function warn before it returns a runaway? At the returned score the far field test has a sensitivity of 0.12, because the walk has already climbed to the bound (Section IX). The seed of the refinement, the best of the 1,000 random candidates of `normalgridtools.cpp:78-88`, is a point the function already holds, and the weighted mean there is one more call. Comparing the bound with that value on the 360 slices of Table II, with the same seeds and a rule written first, would settle whether the warning is worth adding.

What orders the real scrolls, if not the one-sidedness of their segments? The sector ratio explains the one primary loss and fails to order the fifteen, and the two confirmation scrolls where the centroid comes closest are not the one-sided ones (Section VI). A second measure of one-sidedness, declared before it is computed on the fifteen, is what that question needs, not a reading of the present one.

Is there an objective with an interior maximum and without the density bias? Three candidates are real and were not measured. The first is the mean multiplied by a coverage term that counts the normals, and the second a weight chosen so that the mean itself decays far from the section. The third is a consensus set that rejects a segment whose normal is wrong instead of averaging it in, as the header comment of the function announces. The seed score of `normalgridtools.cpp:78-88` and the refinement score of `normalgridtools.cpp:98-117` are also two different functions, so the search can start the refinement from a point its own objective would never have chosen. The exponent belongs here as well: 0.5 would be preferred by a reader who puts the density bias ahead of accuracy (Section VIII-A). Each of these is a new objective or a new search. It would be settled on this bench, on the fifteen scrolls and the synthetic sections of Table VI, with the order of the criteria declared before the run.

XI. Conclusion

The refinement score of the estimator divides by the sum of its weights, which makes it a weighted mean whose limit far from the section is the largest eigenvalue of the covariance of the normals. On a partial prediction that limit can exceed the score on the axis, and there the hill climb is right to walk out. It walks out on 135 of 360 slices with a reference, and 134 of those departures satisfy that condition, the remaining one, on PHerc. 0257 z 8479, sitting at the majority line under the shipped walk. Without the division the estimate lands inside the volume on 576 of 576 slices of 24 scrolls, against 350 as published. The criteria that run also registered on the stability between seeds and on the continuity along z fail, so it does not show that the change generalises. The price is a bias towards the denser side of a section, 0.44 to 2.44 mm on synthetic sections whose centre is known by construction. What is not established is that the weighted sum is the correct geometric objective, or that the change generalises. Nor is it established how its distances compare with the error of a hand annotation, which is not measurable on today's public data.

For someone using the code, the change is one line and it is in the main branch of `villa`. The estimator no longer

¹⁷`inputs/external-references/PROVENANCE.md`.

¹⁸Read from `evidence/second-reference.csv`, written by `tools/second_reference.py`.

leaves the scan on any scroll we ran it on, and a run can be repeated by passing a seed. The tests keep the known bias from growing unnoticed and fail if the division returns. On a scroll without a reference the estimate is inside the volume, which is not the same as right. On complete concentric sections the published mean remains the more accurate objective, and a reader working on such sections should keep it.

Appendix: where each number comes from

Every file below is in `src/evidence/` and every tool in `src/tools/` of the folder `results/e89eb7ad-umbilicus-estimator/` of the public repository [10], where this article is published. The numbers are printed through macros: `src/paper/rev1_numbers.py` writes those of the revision of 18 and 19 September from these files, `src/tools/paper_numbers.py` those of the second reference and of the state upstream, and the macros of the first run are carried as that run wrote them.

Code and data availability

Everything behind this paper is in `results/e89eb7ad-umbilicus-estimator/` of the public repository [10]: the two pre-registrations, in the original and in English, the evidence files every number is read from, the tools that write them and the sources of this paper. The change is commit `8839ed689` of `villa`, the squash of pull request 1823; before the squash it was commit `7a4129a1d` on the branch `normalgrid-umbilicus-score-not-normalised` of the author's fork. Code is MIT licensed, the paper and its figures are CC BY 4.0, and everything was computed on CPU.

Acknowledgements

`volume-cartographer` is the Vesuvius Challenge's, and so are the five reference umbilici of the primary block, one of them annotated by David Josey. The ten references of the confirmation block are the third party's `herculaneum-umbilici` [7], MIT; the second reference of PHerc. 1218 comes from `vesuvius-sheet-tools` [8], MIT, and is derived from a published set of sheet instance labels as a per slice centroid of the papyrus mask, not clicked. `Umbilicus_Maker` [3] is the manual tool an automatic estimate is meant to spare.

About the author

Giovanni Pellerano is a computer engineer and whistleblowing hacktivist. Co-author and Project Lead of the free and open-source software, `GlobaLeaks`, he is responsible for software development and for the support of the community that has grown around it worldwide. He is a digital rights advocate currently learning more about mainstream generative AI development and how to resist against their various and profound harms.

He had not heard of the Vesuvius Challenge before Saturday 5 September 2026. Everything reported in this series was done

after that date, which is stated because it bounds what the work can be assumed to rest on: no prior familiarity with the data, the tools, the community or the literature, just a curious and hyperfocused mind joining this new journey.

Declaration of generative AI in the research and writing process

During the preparation of this work the author used Claude Fable 5.1 and Claude Opus 5.5 (Anthropic) in order to carry out the full research and to write the article: the models read the data, wrote and ran the tools, performed the measurements, and drafted the text. The author set the direction, supplied the domain judgement, challenged the conclusions, reviewed and edited the content, decided what was published, and takes full responsibility for the content of the publication.

The failure this arrangement is prone to is asserting a conclusion that no measurement supports. It happened repeatedly in the work this paper is cut from. The conclusions that did not survive measurement are not in this paper.

What keeps the arrangement checkable is in the folder and not in this paragraph. The criterion of a study is registered before the run, with the day and the hour it was written, and is not revised once the results are in: the pre-registrations of this article rests on are in `src/prereg/`, with their digests, and in `src/inputs/`. Every number printed here expands a macro computed from a file in `src/evidence/`, and `src/evidence/README.md` names the tool that wrote each file, or says that the tool is not in this folder: no number is typed into the text, by the author or by the model. A reader who doubts a number is not asked to trust the arrangement: the file, the tool and the command that produced it are in the same folder.

References

- [1] Vesuvius Challenge, *Open Problems: Why Reading Every Herculanum Scroll Is Still a Challenge*, 2026, read 30 September 2026. https://scrollprize.org/2026_open_problems
- [2] P. Henderson, "Virtually unrolling the Herculanum papyri by diffeomorphic spiral fitting," *arXiv:2512.04927*, 2025. doi:10.48550/arXiv.2512.04927.
- [3] J. Darby, *Umbilicus_Maker*: manual annotation of scroll umbilici, October 2024. https://github.com/JamesDarby345/Umbilicus_Maker, commit `613b5f16dc9b`.
- [4] Vesuvius Challenge, *villa*: `volume-cartographer` and the scroll processing toolchain, commit `23adee047` of <https://github.com/ScrollPrize/villa>.
- [5] G. Pellerano, "normalgridtools: score the umbilicus with a weighted sum, not a weighted mean," pull request 1823 of <https://github.com/ScrollPrize/villa>, merged as commit `8839ed689`. <https://github.com/ScrollPrize/villa/pull/1823>
- [6] B. A. Nosek, C. R. Ebersole, A. C. DeHaven, D. T. Mellor, "The preregistration revolution," *Proceedings of the National Academy of Sciences*, vol. 115, no. 11, pp. 2600–2606, 2018. doi:10.1073/pnas.1708274114
- [7] A. Drobnikov, *herculaneum-umbilici*: manual umbilicus polylines for ten scrolls, MIT licence, August 2026. <https://github.com/AlexeyDrobnikovStrikesBack/herculaneum-umbilici>, commit `34224b76865c`.
- [8] I. Dopico, *vesuvius-sheet-tools*: sheet instance tools, with an umbilicus for PHerc. 1218 derived from instance labels, MIT licence, July 2026. <https://github.com/IyanDopico/vesuvius-sheet-tools>, commit `cc1fb1429d62`.

TABLE IX

Where each number comes from: the place in the article, what it gives, the file under `src/evidence/` and the tool under `src/tools/` that writes it.
Generated by `src/tools/provenance_rows.py`.

where	what	file	written by
Sec. I	the state upstream of pull request 1823, read again for the introduction	upstream-status-2026-09-28.csv	upstream_status.py
Sec. I	the defect line at the tip of the upstream default branch	upstream-check.csv	--merged-at upstream_check.py
Sec. II, Fig. 2	the score against distance, sum and mean	mechanism-curve.csv	synthetic_density.py
Fig. 3	the same score at each exponent	exponent-ablation-mechanism-curve.csv	figure_exponent.py
Fig. 3	the rate each curve approaches or leaves the plateau	exponent-ablation-cancellation-rate.csv	exponent_ablation.py
Abstract, Sec. II, Table II	the far field bound against the runs, slice by slice	far-field-slices.csv, far-field-prediction.csv	far_field_eigenvalue.py
Sec. II	the far field limit on one slice and two synthetic sections	far-field-eigenvalue.csv	far_field_eigenvalue.py
Table I	the same with the scroll as the unit, and its permutation test	far-field-per-scroll.csv, far-field-permutation.csv	far_field_scrolls.py
Sec. III	which scrolls were measured before the pre-registration	prereg-timeline.csv	prereg_timeline.py
Tables III and IV	distances, intervals and the pre-registered verdicts	cpp-k20.csv, cpp-k20-verdicts.json	cpp_k20.py
Sec. III	every call of the compiled function	cpp-k20-estimates.csv	cpp_k20.py
Sec. III	what those calls cost, summed from the seconds of each call	cpp-k20-estimates.csv	cpp_k20.py
Sec. III, Tables III and IV	the margins before rounding, and the verdicts taken on them	cpp-k20-margins.csv, cpp-k20-verdicts.json	cpp_k20_margins.py
Sec. III	the cap of the transcription's hill climb	search-cap.csv	search_cap.py
Table V	the two rules scored after the fact	baselines-k20.csv	baselines_k20.py
Sec. IV	the same question inside the umbilicus bench	bench-rules.csv	bench_rules.py
Sec. IV	the rank tests and the Bland-Altman rows	inference-refit.json, bland-altman-refit.csv	rev1_inference.py
Abstract, Sec. V, Fig. 1	the run on every scroll of the challenge	cpp-positions-24.csv	cpp_gen24.py
Sec. V	its per scroll summary	cpp-twentyfour.csv	cpp_gen24_score.py
Table VI	the density bias on sections with a known centre	synthetic-density.csv	synthetic_density.py
Sec. VI	one sidedness of the segments on the fifteen scrolls	density-asymmetry-fifteen.csv, density-asymmetry-fifteen-summary.csv	density_asymmetry.py
Table VII	the four cells of the ablation	ablation-search.csv	ablation_search.py
Sec. VIII	what the merged squash of pull request 1823 touches	merged-patch.csv	merged_patch.py
Sec. VIII	the regression test of the patch as built and run	regression-test-run.csv	regression_test_result.py
Table VIII	the exponent of the weight, compiled	cpp-exponent-fifteen-k20.csv, cpp-exponent-seed-spread.csv	cpp_exponent_score.py
Sec. VIII-A	the transcription's $p = 1$ against its own estimates	exponent-ablation-k20-estimates.csv	exponent_ablation.py
Table VIII	the interval and the paired count of its aggregate row	cpp-exponent-aggregate.csv	exponent_intervals.py
Sec. IX	what the second umbilicus of PHerc. 1218 is	second-reference.csv	second_reference.py

- [9] G. Pellerano, *umbilicus-bench*: score a scroll axis estimator against the umbilici the Vesuvius Challenge has published, MIT licence, September 2026. Archived for this article as `src/umbilicus-bench/`, which is the copy its numbers were measured against.
- [10] G. Pellerano, *Vesuvius Challenge: findings, tools and data*, public repository, 2026. <https://github.com/evilaliv3/vesuvius-challenge-pipeline>